ITSOLERA Internship

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Introduction to Booklet

This booklet compiles all the tasks I completed during my summer internship at ITSOLERA.

Each task focuses on practical cybersecurity skills, including SIEM deployment, firewall configuration, log analysis, and malware research. I've rewritten the content in my own words to reflect my personal understanding and experiences, ensuring originality.

The tasks build on each other, starting with basic tool setups and progressing to advanced analysis and reporting.

The booklet also includes the Malware Breach Report (Task 4), which fulfills by researching and analyzing 10 recent malware-driven breaches.

All work was done in a simulated virtual environment using tools like VMware, Ubuntu, Windows, pfSense, Wazuh, and Suricata. This compilation demonstrates my growth in areas like threat detection, incident response, and security operations.

Task 1: Wazuh Installation and FIM setup

Overview

In this initial assignment, I set up Wazuh, an open-source platform for security monitoring that combines SIEM and XDR capabilities. It helps with tracking logs, aggregating data, and monitoring file changes. Over 10 days, I installed it in a mixed setup with Ubuntu handling the manager side and Windows as the agent. The focus was on enabling File Integrity Monitoring (FIM) to spot unauthorized alterations to files in real time, while tuning it to avoid unnecessary alerts.

Goals

- Install the Wazuh Manager.
- Deploy and link an agent
- Verify basic logging, like Windows login events.
- Configure and test FIM for file creations, edits, and deletions.

Setup Details

- Wazuh Manager: Ran on Ubuntu 22.04 LTS in VMware.
- Version: 4.7.5.
- Agent: Installed on Windows 11 (hostname: Window11).
- Manager IP: 192.168.189.131.

Installation Steps

I started by preparing Ubuntu: I ran updates with **sudo apt update && sudo apt upgrade -y** ,to get the latest packages.

Then, I added dependencies like **curl** and **gnupg** using **sudo apt install curl unzip apt-transport-https lsb-release gnupg -y**.

Next, I downloaded the installation script from the official site: **curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh,** made it executable with **chmod +x wazuh-install.sh**, and ran it with **sudo bash ./wazuh-install.sh -a**. This took about 15 minutes and set up the dashboard, accessible via the Ubuntu IP with provided credentials.

For the agent, I used the Wazuh dashboard to generate deployment instructions for Windows 11. I ran the PowerShell script it provided, which installed the agent and connected it to the manager.

Configuring FIM

FIM (**File Integrity Monitoring)** checks file checksums to detect **changes**, **creations**, or **removals**.
It's enabled by default on the agent, but I customized it in the ossec.conf file. I added directories like:

C:\Users\thebe\Desktop, **C:\Users\thebe\Documents**, **C:\Users\Public**

using the <directories> tag, along with check_all (to monitor everything) and realtime (for instant alerts, even if reports are scheduled every 12 hours).
After restarting the agent, I tested by creating, modifying, and deleting files in those folders.
The Wazuh dashboard on Ubuntu showed real-time logs for these actions.

FIM

Task 2: Integrating Wazuh with pfSense and Setting Up Firewall Rules

Overview

For this task, I built a secure virtual network where pfSense acts as a software firewall, controlling traffic and integrating with Wazuh for monitoring. I set rules to block based on locations, domains, and access levels, then tested everything in a lab setup.

Goals

- Deploy pfSense in VMware.
- Create rules for geolocation, domain, and admin access blocking.
- Link pfSense to Wazuh via Syslog for live alerts.
- Analyze traffic from logs

Setup Details

- Ubuntu 22.04: Hosts Wazuh Manager and acts as an internal workstation (IP: 192.168.213.130 on VMnet1 – Host Only).
- pfSense Firewall: LAN IP: 192.168.213.1 (VMnet1); WAN IP: 192.168.18.130 (Bridged for internet).
- Network: VMnet1 for internal traffic; Bridged for external. Ubuntu routes through pfSense as gateway.

Topology

Ubuntu 22.04 / Wazuh Manager
192.168.213.130
| |
| |
VMnet1 (Host Only)
| |
| |
LAN - pfSense Firewall
192.168.213.1
| |
| |
WAN - pfSense Firewall
192.168.18.130
| |
| |
Internet (Bridged)

Firewall Choice and Setup

I picked **pfSense** for its strong features like stateful inspection and **GeoIP** support. I installed **pfBlockerNG-devel** for advanced blocking and linked it to **MaxMind GeoLite2**

Rule Configurations and Tests

Geolocation Blocking (Russia & China): To reduce threats from risky areas, I enabled GeoIP in pfBlockerNG, created aliases for RU and CN, and applied inbound/outbound blocks on WAN/LAN. Tested by trying sites from those regions – access denied, logged in pfSense and Wazuh.

Wazuh Integration

I configured pfSense to send Syslog to Ubuntu's Wazuh Manager, which parsed the logs using built-in decoders.

Traffic Insights

Geo-blocks stopped many unwanted TCP connections from risky countries. Domain rules prevented access without workarounds.

Benefits and Conclusion

This setup shrinks attack surfaces, enforces policies, and centralizes monitoring – a solid defense-in-depth approach for SOCs.

Task 3: Analyzing a Simulated Malware Download Incident

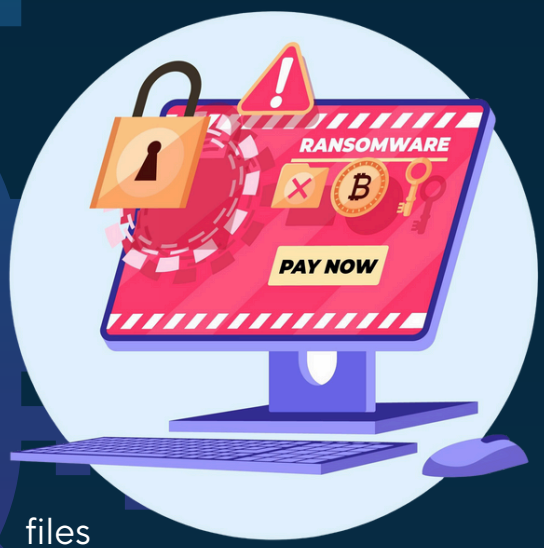
Overview

I simulated downloading WannaCry ransomware from theZoo GitHub repo to study logs from pfSense (via Suricata) and Wazuh. The zip file (SHA256: ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa, password: infected) was cloned but not executed. Detections flagged network and file activity.

No real infection happened, but tools caught repo access and file changes. IOCs include hashes and URLs; IOAs cover download behaviors.

Log Review

- pfSense/Suricata: Alerts for TLS SNI to github.com/theZoo (e.g., "**SURICATA TheZoo GitHub TLS SNI Access**" on 09/03/2025). Also noted stream anomalies, but no blocks.
- Wazuh FIM: File additions/modifications in /home/thebe562/Downloads/theZoo/ (e.g., git hooks and packs, Rule IDs 554/550/550 on Sep 3, 2025).



Malware Details

WannaCry spreads via SMB exploits, encrypts files (.WNCRY extension), and demands ransom. Here, it stayed zipped.

IOCs and IOAs

- IOCs: Hashes (MD5: 84c82835a5d21bbcf75a61706d8ab549), File: Ransomware.WannaCry.zip, URL: Your paragraph text.
- IOAs: Git clone from malware repo, mass file adds

Recommendations

Block malware domains in Suricata; auto-quarantine in Wazuh; review FIM regularly.

Incident Response Plan (NIST-Based)

Preparation

Form a Computer Security Incident Response Team (CSIRT) with key roles like Incident Coordinator and Technical Analyst. Equip the team with tools such as Suricata for network monitoring and Wazuh for endpoint detection, and establish policies like avoiding malware execution outside VMs. Conduct regular training and set up baseline monitoring for threats like WannaCry hashes.

Detection/Analysis

Identify incidents using Suricata logs, Wazuh alerts, and user reports. Triage severity, collect logs and hashes, and analyze Indicators of Compromise (IOCs) and Attack (IOAs) with tools like VirusTotal. Document findings for affected systems.

Containment

Isolate affected systems using pfSense rules and quarantine suspicious files like Ransomware.WannaCry.zip. Block risky domains and apply patches (e.g., MS17-010), while preserving evidence with disk images.

Eradication

Remove malware by deleting files and scanning remnants with ClamAV. Fix vulnerabilities with stricter download controls and update security settings to prevent recurrence.

Recovery

Restore systems from clean backups after scans, validate operations, and monitor for 48 hours. Notify stakeholders if needed and reintegrate systems cautiously.

Task 4: Malware Breach Report

Executive Summary

This report examines 10 significant recent malware breaches (2023-2025) in sectors like healthcare, finance, cloud, hospitality, telecom, and government. For each, I cover the incident, attack method (vector/malware/actor), impact, and consequences. Drawn from credible sources like CISA, Reuters, KrebsOnSecurity, and Wired.

Methodology

Selected publicly documented incidents involving malware/ransomware/data theft, deemed "major" in coverage. Sources: CISA advisories, AP/Reuters, specialist journalism. Citations per section. 10 such incidents have been collected among which some are listed below:

1)Change Healthcare / UnitedHealth (Ransomware/Data Exfiltration - Feb 2024; Disclosures 2024-25)

What Happened: Hit by ALPHV/BlackCat; disrupted claims, exfiltrated patient data. Vector: Weak remote access (no MFA), credentials for persistence, double-extortion.

Impact: ~100M+ affected; delays in claims, billions in costs. Aftermath: Highlighted third-party risks; enforce MFA/least privilege.

Citations: CISA, AP News, BlackFog, Healthcare Dive.

2. Snowflake Customer Compromises (Credential-Based Data Theft/Extortion - Mid-2024)

What Happened: Stolen credentials targeted Snowflake instances; data theft from customers like Ticketmaster/AT&T, with extortion.

Vector: Credential stuffing/reuse, no MFA; post-compromise access. Impact: 100+ customers, millions of records exposed; arrests followed. Aftermath: Stress MFA/rotation, anomaly detection.

Citations: WIRED, Snowflake, Financial Times, Google Cloud.

3. MOVEit (Cl0p Exploitation of MFT Vulnerability - 2023, Disclosures 2024)

What Happened: CL0P exploited CVE-2023-34362 in MOVEit for file theft/extortion across sectors.
Vector: Zero-day web-app vuln allowing unauth exfiltration.
Impact: Thousands of orgs, tens of millions records; supply-chain cascade.
Aftermath: Patch urgently, IOC hunts, vendor comms.
Citations: CISA, konbriefing.com, CISA

4. MGM Resorts (Operational Ransomware/Social Engineering - Sep 2023)

What Happened: Social engineering disrupted ops (reservations/slots); ALPHV/Scattered Spider stole data.
Vector: Vishing for creds, lateral movement to ransomware.
Impact: 10-day downtime, millions records; financial hits, lawsuits.
Aftermath: Train staff, secure support auth, continuity plans.
Citations: MGM Investors, 1Password Blog, AP News, TraceSecurity.